GreenPower

ZW□-40.5 Series

Outdoor High Voltage AC Vacuum Circuit Breaker





About Us»

GreenPower is a manufacturer of medium- and low-voltage switchgear and components, with in-house engineering, manufacturing and testing capabilities. Our manufacturing entity, Zhejiang GreenPower Electric Co., Ltd., was established in 1992. International business and export operations are handled by Zhejiang GreenPower I&E Co., Ltd.

Our product portfolio covers medium-voltage switchgear from 12 kV to 40.5 kV, including air-insulated metal-clad switchgear (AIS), cubicle-type gas-insulated switchgear (C-GIS), ring main units (RMU), vacuum circuit breakers, and a broad range of switchgear components and accessories. We also manufacture low-voltage withdrawable switchgear systems with ratings up to 6300 A.

Engineering and manufacturing are central to what we do. Our production and testing capabilities include automated sheet-metal processing, CNC laser cutting, 3D five-axis laser welding for stainless-steel gas tanks, automatic helium leak detection, and a fully shielded partial-discharge test room. These capabilities support consistent manufacturing quality from enclosure fabrication through assembly and final testing.

For customers moving toward lower-impact power distribution systems, GreenPower offers SF6-free solutions including dry-air insulated RMUs and nitrogen-insulated C-GIS, alongside conventional gas-insulated solutions where required. These technologies help customers respond to increasingly stringent environmental requirements for medium-voltage switchgear.

We also support project-specific engineering based on customer drawings and specifications, including non-standard dimensions, special voltage and frequency requirements, high-altitude applications, customized component selection and other application-specific configurations.

Our goal is simple: to be the switchgear manufacturer our customers return to — because the equipment is properly engineered, reliably manufactured and built for long-term service.

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1. Product overview and use

ZW □ - 40.5 series outdoor high voltage AC vacuum circuit breaker is an outdoor switch equipment with rated voltage of 40.5kV and frequency of 50Hz. It is mainly used in three-phase AC power systems, 33~40.5kV voltage level substations and outdoor lines, to open and close. Load current, overload current, short-circuit current, capacitive current, inductive current in transmission and distribution lines, as the core equipment for power control and protection of power transmission and distribution substations and lines;

This equipment has the advantages of high breaking capacity, excellent insulation performance, high frequency operation reliability, long service life, and low maintenance.

This equipment is suitable for industrial and mining, enterprises, power plants, substations, overhead lines, and places where various loads of different natures are switched and frequently operated; it is also suitable for double power connections, cutting no-load long lines, and opening and closing capacitor banks. For the protection and control of electrical facilities.

This equipment complies with GB 1984-2014 "High Voltage AC Circuit Breaker" and IEC 62271-100:2008 "High Voltage AC Circuit Breaker" and other national and international standards.

This equipment has four parts: ZW □ - 40.5 switch body, voltage transformer, FTU fault detection controller, meter box, etc. The four parts are electrically connected through aviation sockets and outdoor sealed control cables, which can form a complete set of intelligent switch for the circuit The equipment ZW □ - 40.5JLFK" meets the needs of customers such as centralized or on-site relay protection, power metering, payment management, and three remotes.

This equipment, "Complete set of intelligent switchgear ZW □ - 40.5JLFK", has high-voltage power metering, pre-paid tripping and power failure, payment and closing power transmission, fault detection, protection control and communication functions; it can reliably judge and detect inside and outside The milliamp zero sequence current, as well as the phase-to-phase over-current and short-circuit fault currents; realize automatic removal of single-phase earth faults, phase-to-phase over-current, short-circuit faults, and uninterrupted power demand for reclosing.

The default operating environment of this device is within 1000 meters above sea level.

This equipment can be customized according to the use environment, 2000 meters, 3000 meters, 4000 meters, 5000 meters above sea level, 4 specifications of products.

There are 2 kinds of insulation materials for the external insulation of this equipment, such as porcelain bottle and silicone rubber.

ZW7A-40.5 type:

CT is built-in or externally, and the external insulation is the physical appearance of silicone rubber or porcelain.



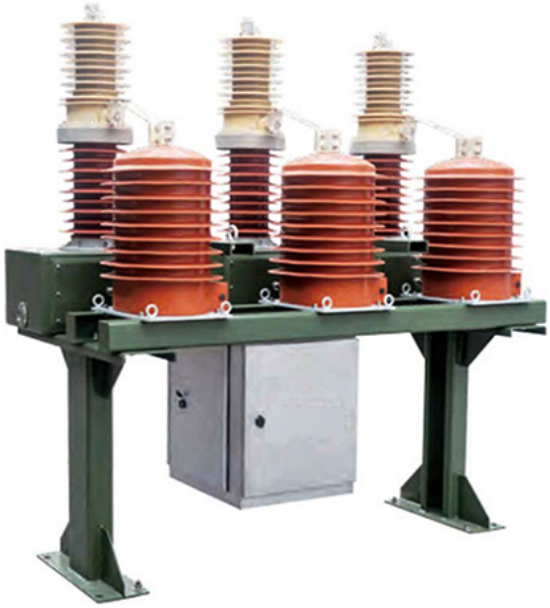
ZW7B-40.5 type:

CT is externally mounted, and the external insulation is the physical appearance of silicone rubber or porcelain.



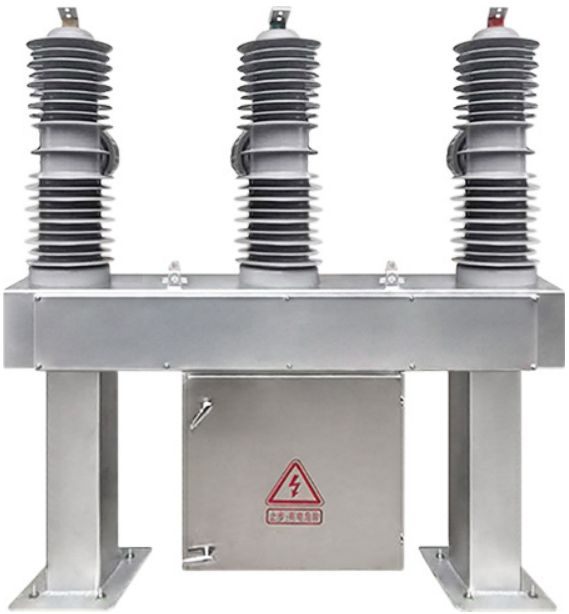
ZW7C-40.5 type:

Embedded pole, external CT, external insulation is the physical appearance of silicon rubber or porcelain.



ZW7D-40.5 type:

Embedded pole, CT built-in or external, external insulation is the physical appearance of silicon rubber.



ZW7E-40.5 type:

Embedded pole, CT built-in or external, external insulation is the physical appearance of silicon rubber.



2. Product model and meaning

Z W □ - 40.5 / T 1250 - 25 JLFK, ZN

- JLFK, ZN — Extended functions: Metering fee control: JLFK, Intelligent model: ZN
- 25 — Rated short-circuit breaking current (kA)
- 1250 — Rated current (A)
- T — Operating mechanism: T spring, M permanent magnet
- 40.5 — Rated voltage (kV)
- □ — Design serial number: 7A, 7B, 7C, 7D, 7E
- W — Conditions of use: Outdoor
- Z — Arc extinguishing medium: Vacuum

3. Reference standard

GB l984 "High Voltage AC Circuit Breaker"

GB/T ll022 "Common Technical Requirements for High Voltage Switchgear and Control Equipment Standards"

GB/T 311.2 "Insulation Coordination"

GB/T 4109 "Insulating Bushing with AC Voltage Higher than 1000V"

GB/T 4208 "Shell Protection Grade"

GB/T 16927.1 "High Voltage Test Technology"

GB 2706 "Test Method for Dynamic Thermal Stability of AC High Voltage Electrical Appliances"

GB 3309 "High Voltage Switchgear Mechanical Test at Room Temperature"

GB/T 4585 "Artificial Pollution Test of High Voltage Insulators for AC System"

GB/T 7354 "Partial Discharge Measurement"

DL/T 403 "10~35kV Indoor High Voltage Vacuum Circuit Breaker Ordering Technical Conditions"

DL/T 593 "Common Ordering Technical Conditions for High Voltage Switchgear"

DL/T 402 "AC High Voltage Circuit Breaker Ordering Technical Conditions"

IEC 62271-100

Remarks: The above standards, quote the latest years.

4. Working conditions

Normal use conditions

a. Ambient air temperature, minimum -30°, maximum +40°; the average temperature measured within 24 hours does not exceed 35°;

b. The altitude does not exceed 1000m;

c. The pollution level of the surrounding air shall not exceed level II;

d. The surrounding air is not obviously polluted by dust, smoke, corrosive or flammable gas, steam or salt spray;

e. The thickness of icing shall not exceed 20mm;

f. The wind speed is not more than 34m/s;

g. The seismic fissure does not exceed 8 degrees;

h. Humidity conditions:
The average value of relative humidity measured within 24h, not exceeding 95%;
The average value of the water vapor pressure measured within 24h does not exceed 2.2kPa;
The average monthly relative humidity does not exceed 90%;
The average monthly water vapor pressure does not exceed 1.8kPa;

If the above-mentioned normal use conditions are exceeded, please consult with the manufacturer in advance, and please specify when ordering;

Abnormal use conditions

Including the extreme use environment such as altitude above 1000 meters, rapid temperature change, icing above 20mm, heavy pollution, severe condensation, mildew, sand, dust, severe cold, scorching heat, vibration, impact, swing, etc., please negotiate in advance when ordering.

5. Main technical parameters

Table of main technical parameters

No.	Item	Unit	Data		
1	Rated voltage	kV	40.5		
2	Rated power frequency withstand voltage 1min	kV	95		
3	Rated lightning impulse withstand voltage	kV	185		
4	Rated frequency	Hz	50		
5	Rated current	A	630, 1250, 1600, 2000, 2500		
6	Rated short-time withstand current	kA	20	25	31.5
7	Rated peak withstand current	kA	50	63	80
8	Rated short-circuit breaking current	kA	20	25	31.5
9	Rated short-circuit making current	kA	50	63	80
10	Rated short-circuit duration	s	4		
11	DC component of rated short-circuit breaking current		51		
12	Peak value of transient recovery voltage (TRV)	kV	114		
13	Rated power supply voltage of closing and opening devices and auxiliary circuits	V	DC/AC 220V, DC/AC 110V		
14	Rated operating sequence		O-0.3s-CO-15s-CO		
15	Opening time	ms	20 ~ 50		
16	Closing time	ms	30 ~ 80		
17	DC time constant of rated short-circuit breaking current	ms	45		
18	Rated cable charging breaking current	A	50		
19	Service life		E2-C2-M2 (10000)		
20	Rated short-circuit breaking current breaking times	times	20		

Table of assembly adjustment parameter

No.	Item	Unit	Data
1	Contact opening distance	mm	20 ± 2
2	Contact stroke	mm	4 ± 1
3	Average closing speed (measurement contact is just before closing to 10mm before closing)	m/s	0.8 ± 0.3
4	Average opening speed (measurement contact is just divided to 10mm apart)	m/s	1.6 ± 0.3
5	Bounce time of contact closing	ms	≤ 5
6	Bounce time of contact closing	ms	≤ 2
7	Three-pole contact opening in different periods	ms	≤ 2
8	Main circuit resistance of each pole	μΩ	≤ 100 (Without instrument transformer)

Note: The above parameters refer to the rated operating voltage

6. Product structure characteristics

The structure of the circuit breaker is a porcelain bottle pillar type, the upper porcelain bottle is an arc extinguishing chamber porcelain bottle, and a vacuum interrupter is installed inside, and the lower porcelain bottle is a pillar porcelain bottle. The three-phase porcelain bottles are installed in a structural box. The moving end of the arc extinguishing chamber is connected with the output shaft of the spring mechanism through a crank arm and an insulating pull rod. The link connecting the spring mechanism and the circuit breaker is installed in a waterproof mechanism box.

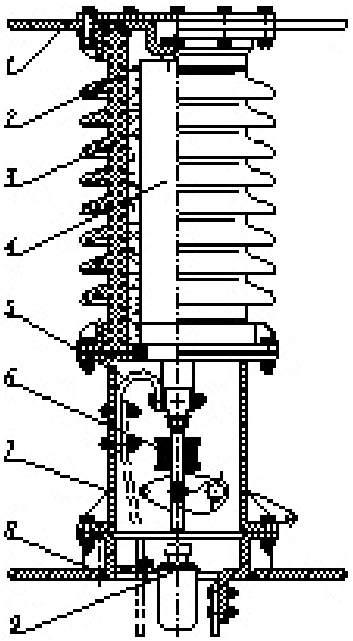
This product is mainly composed of vacuum interrupter, current transformer, transmission mechanism, operating mechanism, mechanism box, structure box, etc.

Vacuum interrupter (VI)

The circuit breaker is equipped with a VI, which has a very high degree of vacuum. Because the contact of the interrupter is made of CuCr50 material and adopts a cup-shaped longitudinal magnetic field structure, it has large breaking capacity, high insulation level and strong arc extinguishing ability. , Long life, small size, etc. and the advantages of simple maintenance, no explosion hazard, no pollution, and low noise.

The vacuum interrupter is the main component of the circuit breaker, and the structure of the interrupter assembly is as shown in the right figure:

No.	Description
1	The upper outlet terminal
2	Porcelain bottle
3	Vacuum insulating silicone grease
4	Vacuum interrupter
5	Soft connection
6	Contact spring
7	Upper bracket
8	Lower outlet terminal
9	Oil buffer



ZW7A-40.5 type, ZW7B-40.5 type

The structure of the arc extinguishing chamber assembly is shown as follows:

The VI is installed in the upper porcelain bottle, which not only avoids the many disadvantages caused by oil filling and SF6 gas filling, makes the inspection and maintenance more convenient. Between the interrupter and the porcelain bottle, it is filled with insulating material silicone between the interrupter and the porcelain bottle and solve the condensation problem.

The insulating material has excellent characteristics such as good insulation performance, aging resistance, ultraviolet resistance, high temperature resistance, low temperature resistance, sealing, lubrication, inertness, non-volatile, and strong water repellency.

ZW7C-40.5 type, ZW7D-40.5 type, ZW7E-40.5 type

The structure of the arc extinguishing chamber assembly is shown as follows:

The solid-sealed pole (interrupter assembly) is directly sealed in epoxy resin through the advanced automatic pressure gel process (APG). It simplifies the assembly process of the pole, prevents the connection bolts of the vacuum interrupter conductive circuit from loosening due to vibration in operation, improves reliability, and makes the outer surface of the vacuum interrupter free from external mechanical forces and external environment (The influence of dust, moisture, pollution, high altitude, small animals) further improves the electric field distribution of the pole.

For the thermal expansion coefficient of the vacuum interrupter ceramic shell, there is a big difference (1:10) from the expansion coefficient of the epoxy resin casting insulation layer. In order to avoid the circuit breaker from solidifying the pole, during the manufacturing and transportation process, due to environmental temperature changes The problem of cracking of the epoxy resin casting insulation layer is caused by a layer of high dielectric strength, high air tightness, and high elasticity between the surface of the ultra-low resistance vacuum interrupter and the epoxy resin casting insulation layer. The imported buffer layer is used to absorb the mutual squeezing force between two materials with different thermal expansion coefficients, thus thoroughly ensuring that the epoxy resin casting solid insulation layer will not crack under the changing conditions of high and low temperature.

Current transformer

The circuit breaker can be equipped with a maximum of three current transformers, and each current transformer can include a maximum of four coils, of which one is used to measure 0.2S level, one is used to measure 0.5 level, and two are used to protect P level. The current conversion ratio can be selected according to user requirements, ranging from 25-2000/5, and the multiple conversion ratio multiples are 1 times, 2 times, and 3 times. The current transformer has the characteristics of high precision and large output capacity.

Examples of typical parameters:

A. 00-200-300/5; 0.2S/0.5/5P20; 10/20/10 VA

B. 100-200-300/5; 0.2S/0.5/5P20/5P20; 10/20/10/10 VA

C. 100-200-300/5; 0.2S/0.5/10P10; 10/20/25VA

D. 100-200-300/5; 0.2S/0.5/10P10/10P10; 10/20/25/25 VA

Specific ordering technical parameters, according to the technical specifications of the project, need to be negotiated and determined!

ZW7A-40.5 type, built-in current transformer

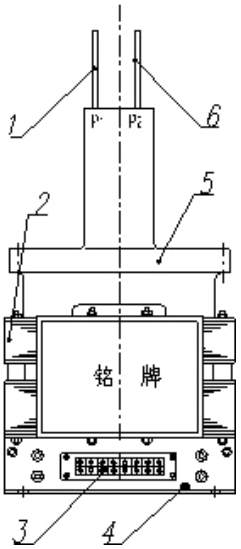
The current transformer is a cast-through type, a fully enclosed structure, and is installed in the lower porcelain bottle.

The current transformer and the lower porcelain bottle are also filled with insulating material silicone to solve the insulation and condensation problems.

This product has a built-in LZZBJ4-35 current transformer, and the shape and structure are shown in the figure:

No.	Description
1	Primary incoming wire terminal
2	Winding
3	Secondary wiring terminal
4	Earthing bolt
5	Connecting flange
6	Primary outgoing terminal





ZW7B-40.5, ZW7C-40.5, ZW7D-40.5, ZW7E-40.5 type, external current transformer

The external current transformer has the characteristics of large creeping distance, large capacity, silicone rubber coating, full insulation, easy maintenance, and easy upgrade!



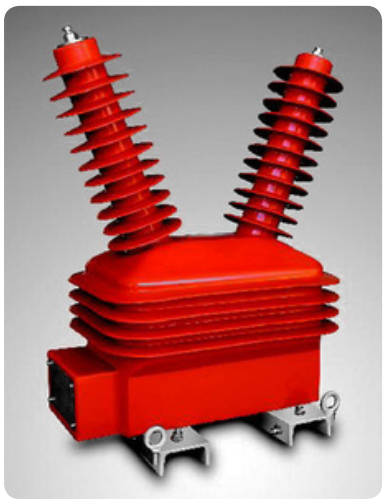


Our factory can design and manufacture special CTs according to user needs.

For example, the accuracy level is 10P20, 5P20, 0.2S, 0.1; the rated load impedance is increased; the rated secondary current is 1A and other special parameters. Various specific technical parameters can be resolved through negotiation.

Voltage transformer

The external voltage transformer has the characteristics of large creeping distance, large capacity, silicone rubber coating, full insulation, easy maintenance, easy upgrade, etc..



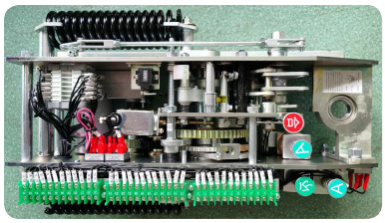
Voltage transformer

Transmission mechanism

Including insulated pull rod, turning arm and guide rod, adjust the length of the insulated pull rod and guide rod, you can adjust the overtravel and contact distance of the circuit breaker.

Operating mechanism

Spring operating mechanism model CT19BW, its parameters, structure and adjustment, please refer to the operating mechanism description in the packing materials.



Operating mechanism

Mechanism box and structure box

The mechanism box and structure box adopt waterproof technology, the door frame adopts waterproof groove, and the door panel adopts foam sealing structure to prevent rainwater from penetrating into the box; the box is equipped with a temperature controller and heater to prevent condensation, resulting in electrical components and mechanical components and parts are damp.



Mechanism box and structure box

Principle of arc extinguishing

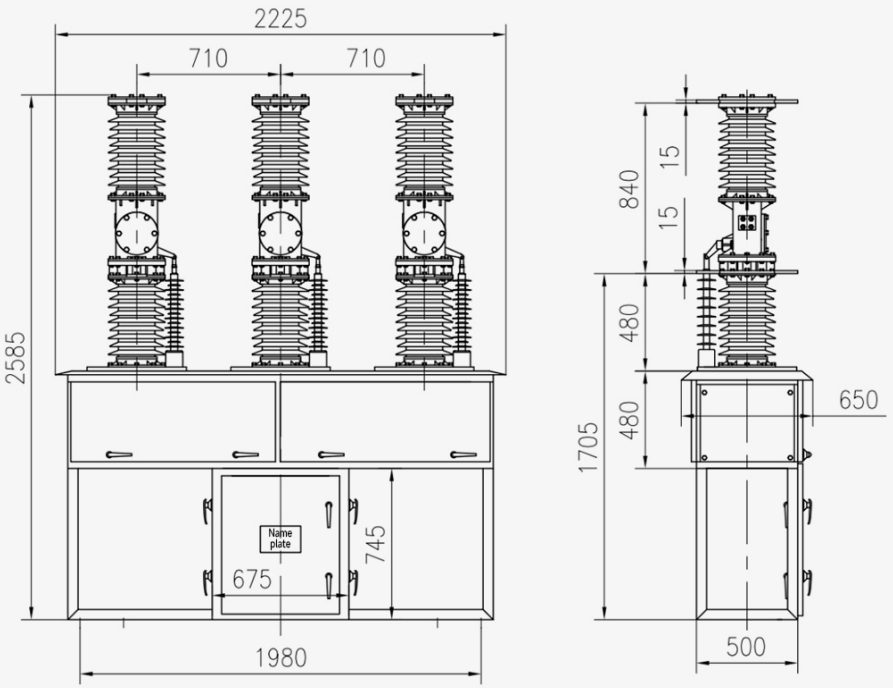
When the moving and static contacts are energized and opened under the action of the operating mechanism, a vacuum arc will be generated between the contacts. At the same time, due to the special structure of the contact, an appropriate longitudinal magnetic field is generated in the contact gap, so that the vacuum arc maintains a diffusion type and maintains a low arc voltage.

When the current naturally crosses zero, the residual ions, electrons and metal vapors can be compounded or condensed on the contact surface and the shield in a microsecond order, and the dielectric strength of the fracture of the arc extinguishing chamber is quickly restored , So that the arc is extinguished, to achieve the purpose of breaking.

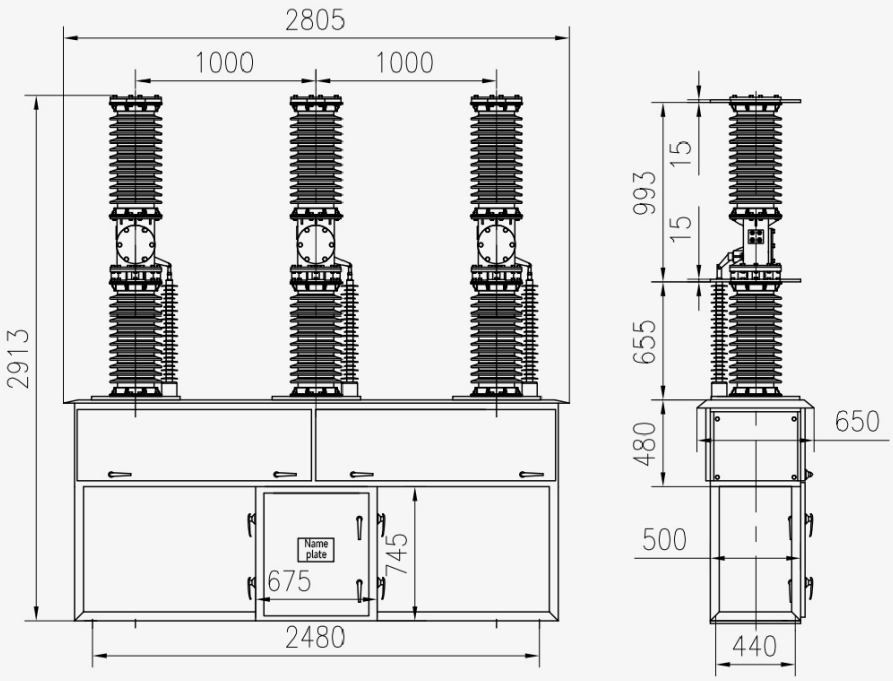
7. Installation dimensions

ZW7A-40.5 type, installation dimension diagram

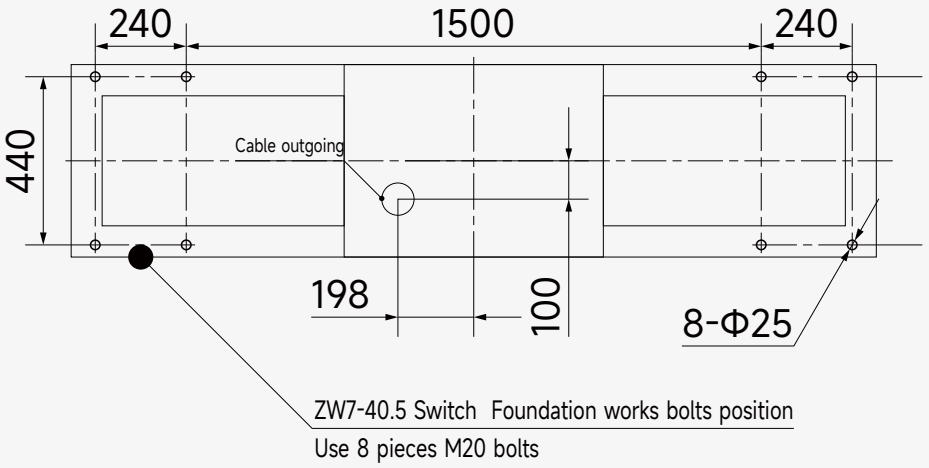
■ **At an altitude of 1000 meters (center distance of 710mm), an altitude of 2000 meters (center distance of 780), and an altitude of 3000 meters (center distance of 850), the overall dimensions of the circuit breaker are as follows:**



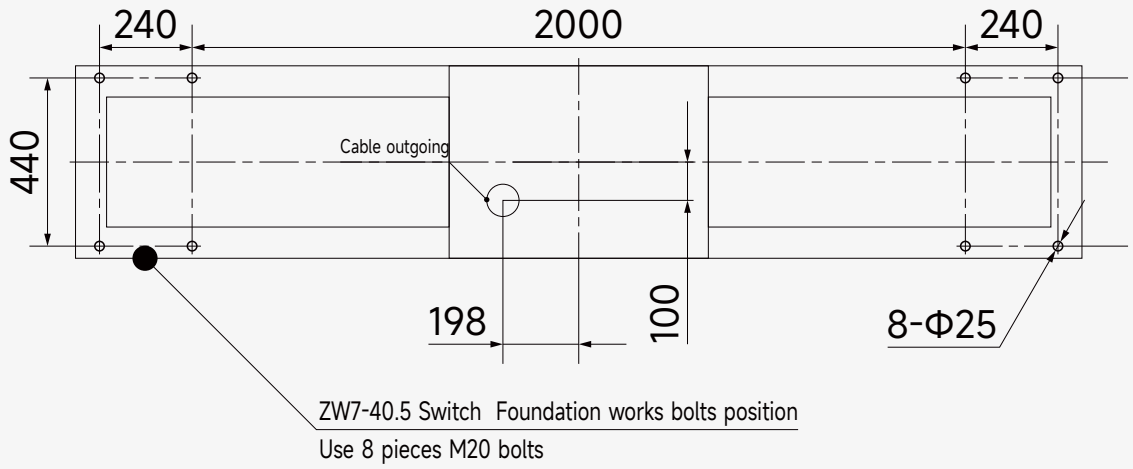
■ **At an altitude of 4000 meters (center distance 920), at 5000 meters (center distance 1000) meters, the overall dimensions of the circuit breaker are as follows:**



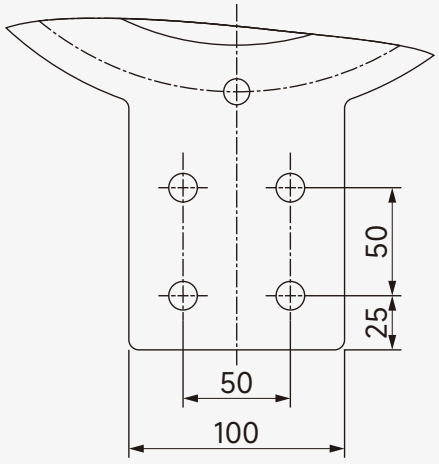
■ **At an altitude of 1000 meters, 2000 meters, 3000 meters above sea level, the installation dimensions of the circuit breaker are as follows:**



■ **At an altitude of 4000 meters above sea level, 5000 meters above sea level, the installation dimensions of the circuit breaker are as follows:**

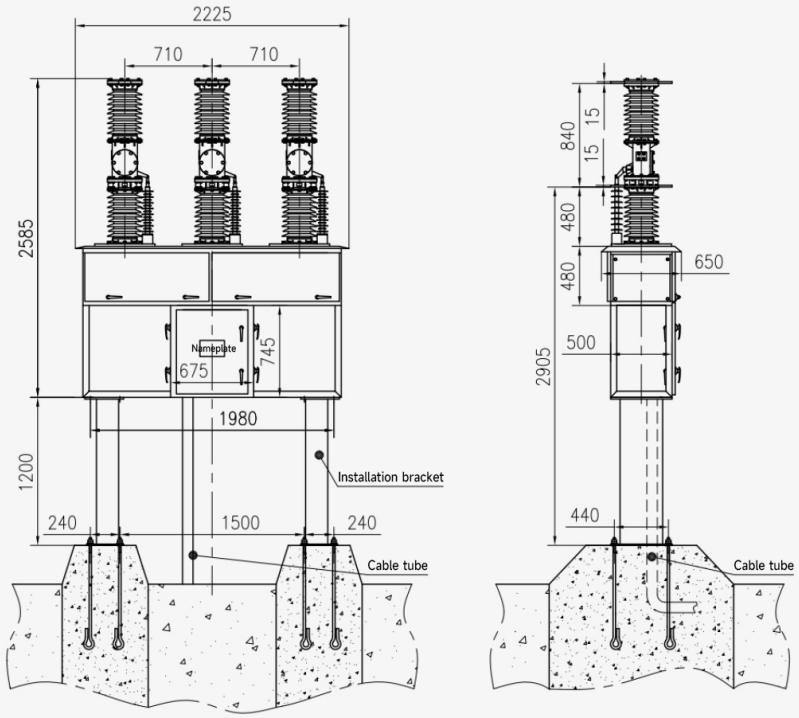


■ **The upper and lower wiring terminals of the circuit breaker are made of aluminum, and the dimensions are as follows:**

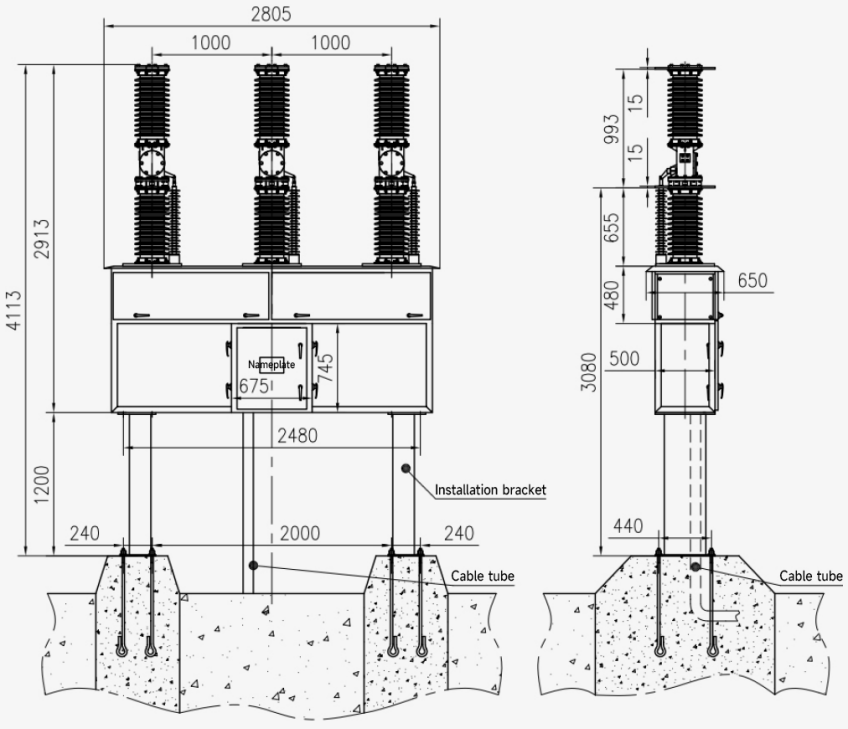


■ **Accurate and professional installation is a basic condition to ensure trouble-free operation of the product.**

a. The altitude is 1000 meters, 2000 meters, 3000 meters, and the basic installation dimensions are the same standard. The following installation methods are recommended:

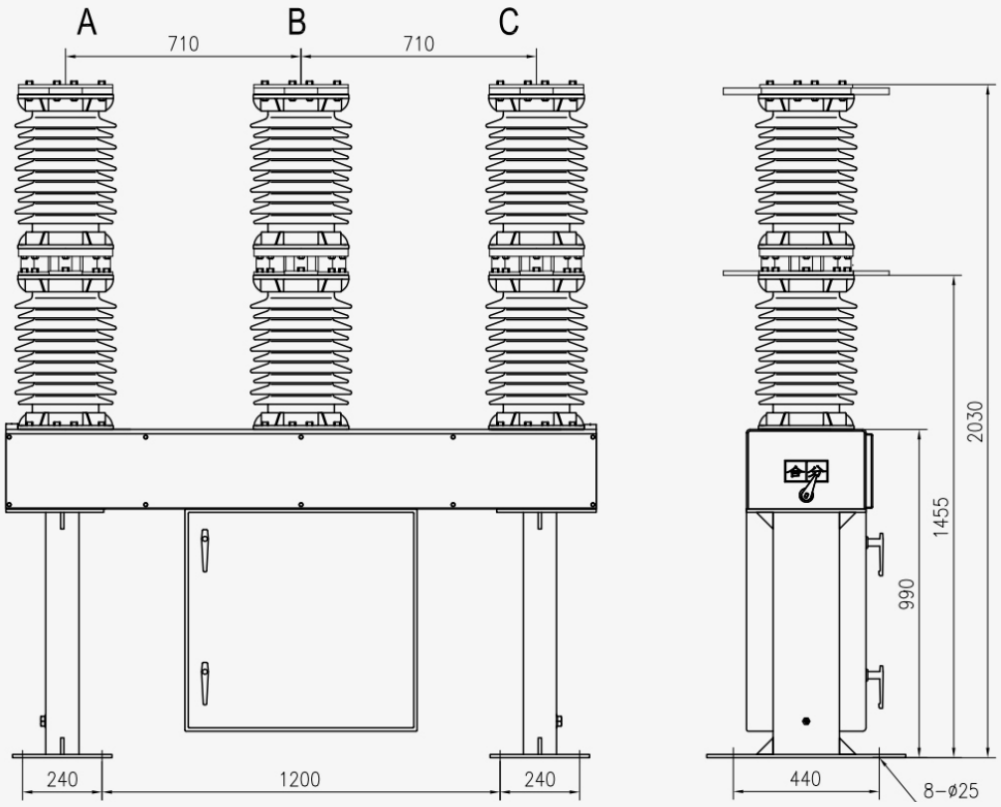


b. The altitude is 4000 meters, 5000 meters, and the basic installation dimensions are the same standard. The following installation methods are recommended:The following installation methods are recommended:

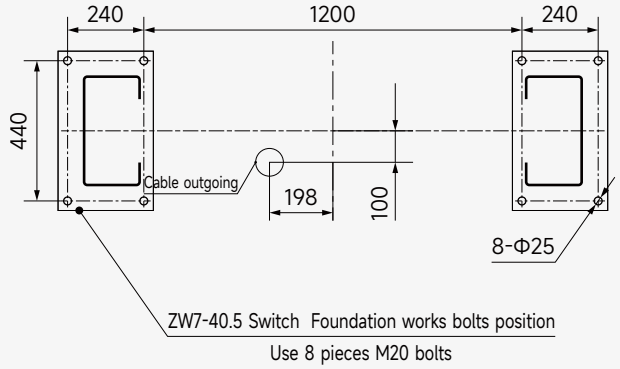


ZW7B-40.5 type, installation dimension diagram

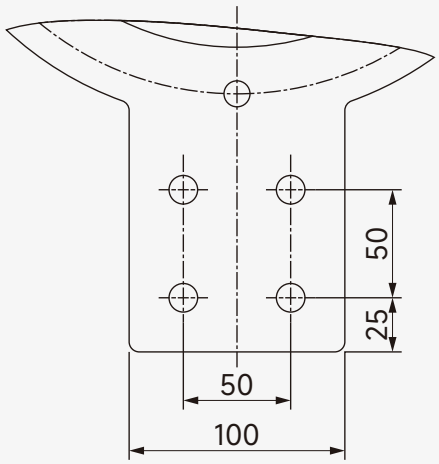
■ At an altitude of 1000 meters and an altitude of 2000 meters, the overall dimensions of the circuit breaker are as follows:



■ At an altitude of 1000 meters above sea level, 2000 meters, the installation dimensions of the circuit breaker are as follows:

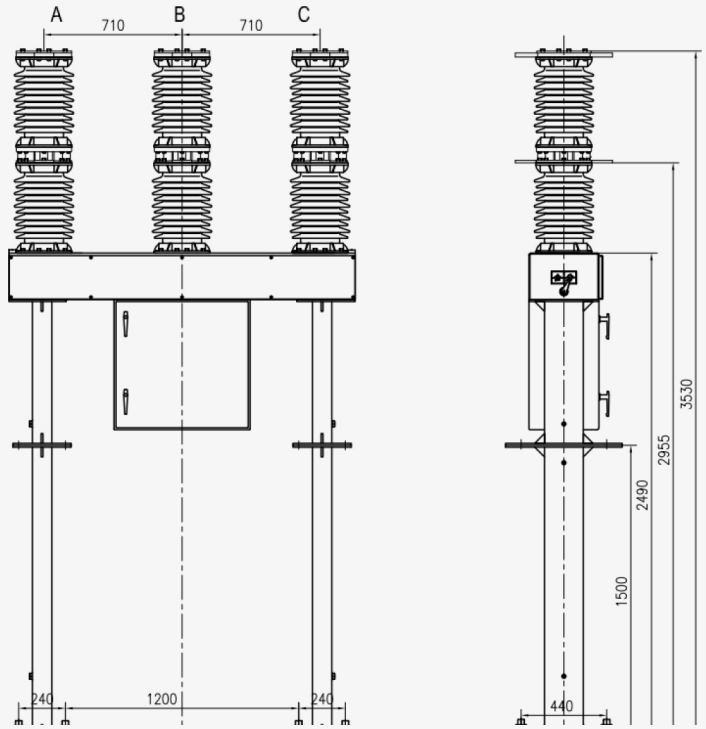


■ The upper and lower wiring terminals of the circuit breaker are made of aluminum, and the dimensions are as follows:



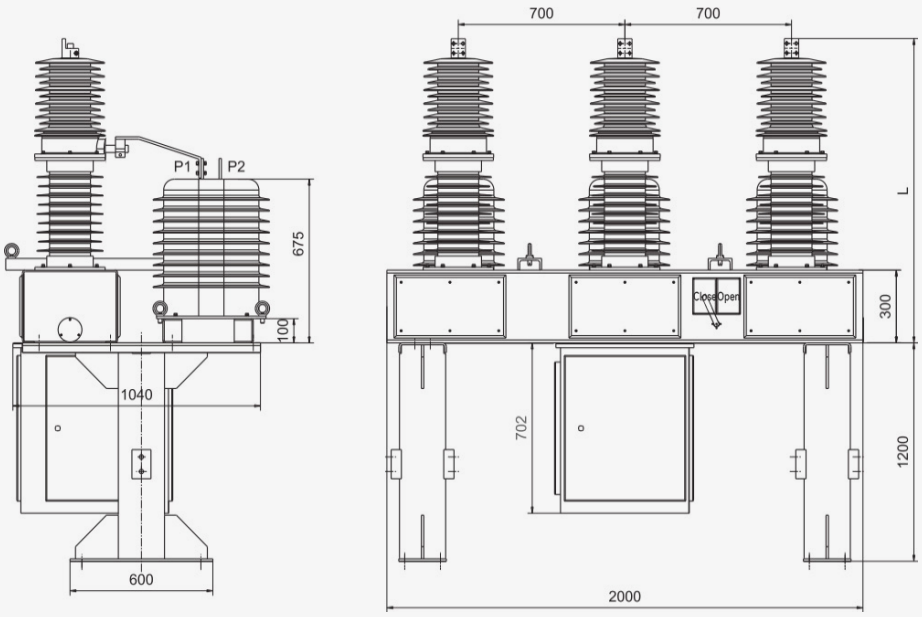
■ **Accurate and professional installation is a basic condition to ensure trouble-free operation of the product.**

The altitude is 1000 meters, 2000 meters, and the basic installation dimensions are the same standard.
The following installation methods are recommended:



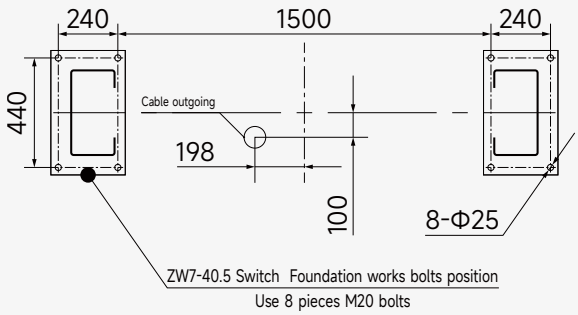
ZW7C-40.5 type, installation dimension diagram

■ **With an altitude of 1000 meters and an altitude of 2000 meters, the overall dimensions of the circuit breaker are as follows:**



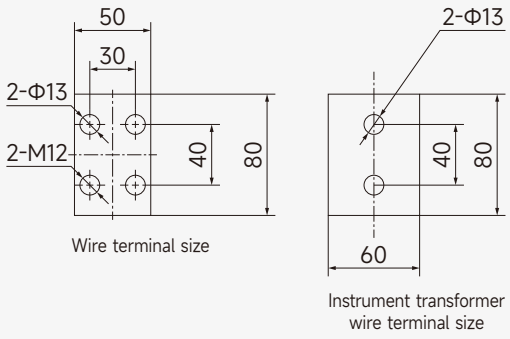
The "L" dimension in the figure changes with the current level, as follows:
ZW37C-40.5/1600-25, L=1640mm ZW37C-40.5/2000-31.5, L=1890mm

■ **At an altitude of 1000 meters above sea level, 2000 meters, the installation dimensions of the circuit breaker are as follows:**

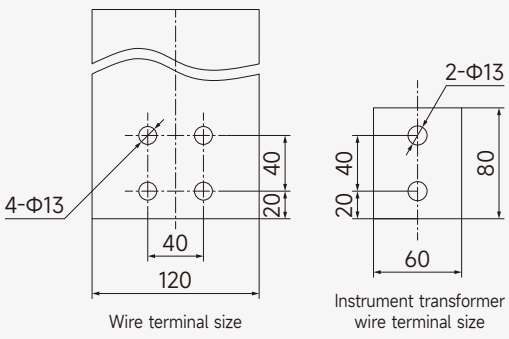


■ **The upper and lower wiring terminals of the circuit breaker are made of copper, and the dimensions are as follows:**

a. The upper and lower wiring terminals of the circuit breaker are 1600A, size drawing

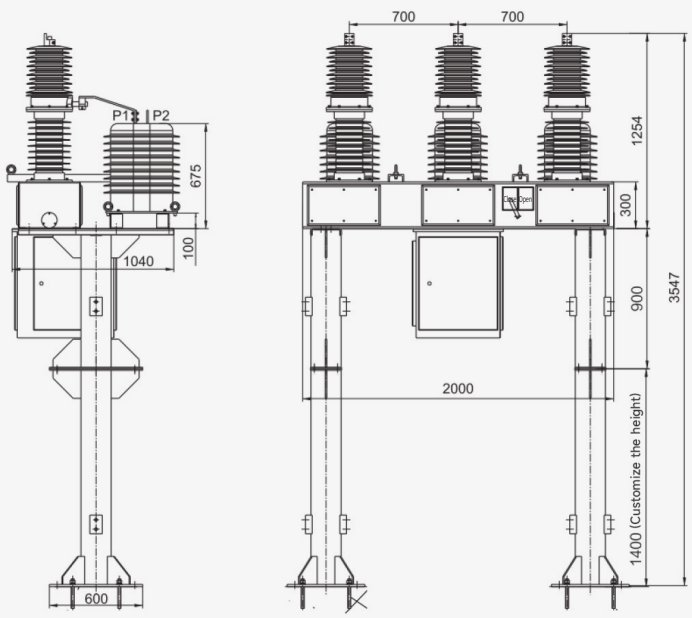


b. The upper and lower wiring terminals of the circuit breaker 2000A, size drawing



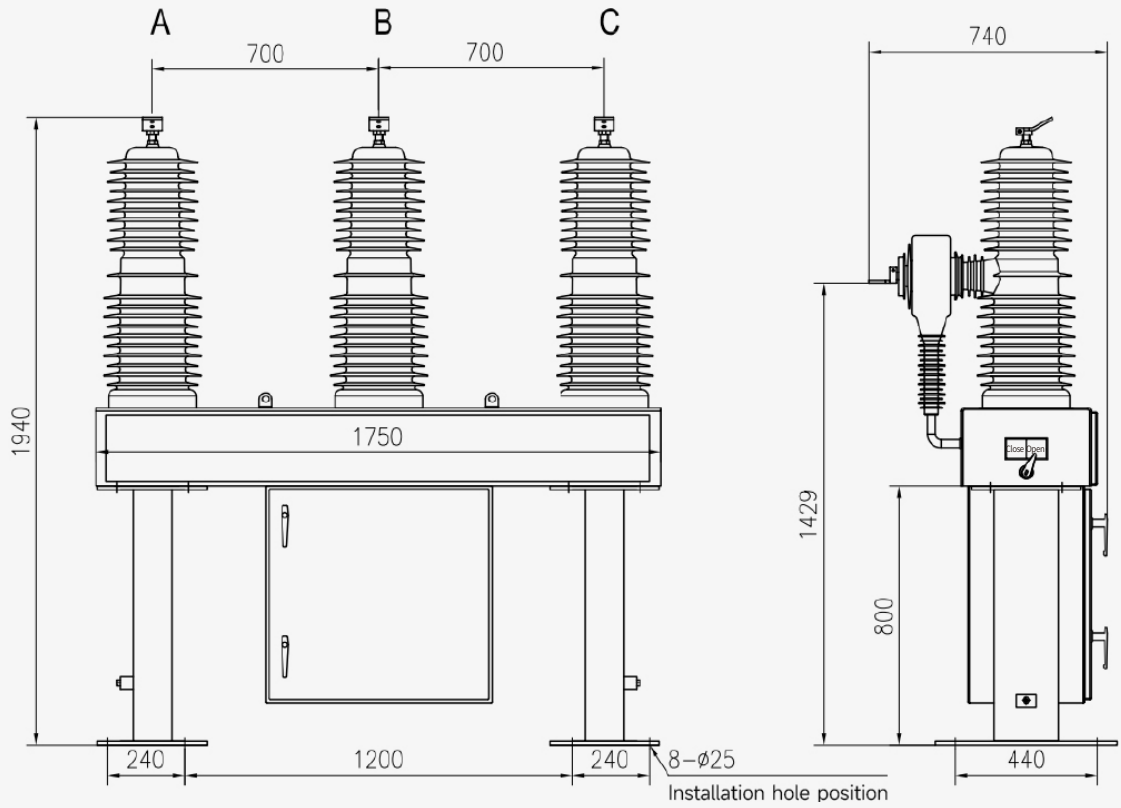
■ **Accurate and professional installation is a basic condition to ensure trouble-free operation of the product.**

The altitude is 1000 meters, 2000 meters, and the basic installation dimensions are the same standard.
The following installation methods are recommended:

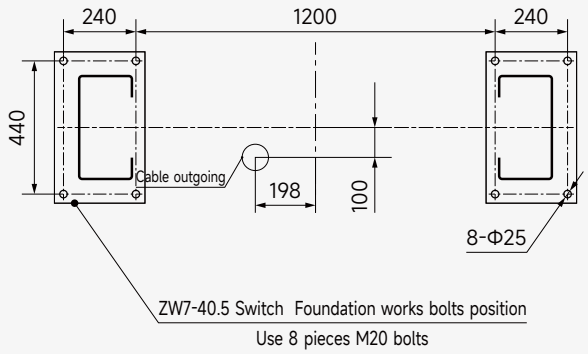


ZW7D-40.5 type, installation dimension diagram

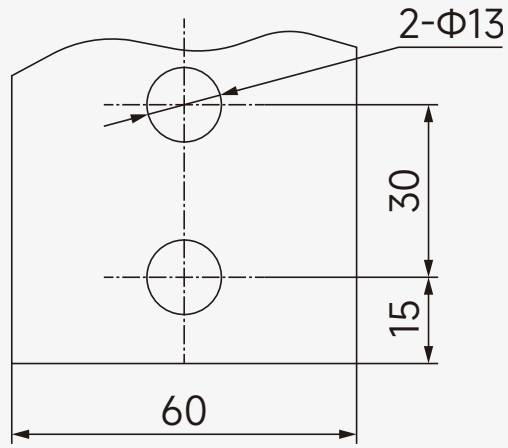
■ At an altitude of 1000 meters and an altitude of 2000 meters, the overall dimensions of the circuit breaker are as follows:



■ At an altitude of 1000 meters above sea level, 2000 meters, the installation dimensions of the circuit breaker are as follows:

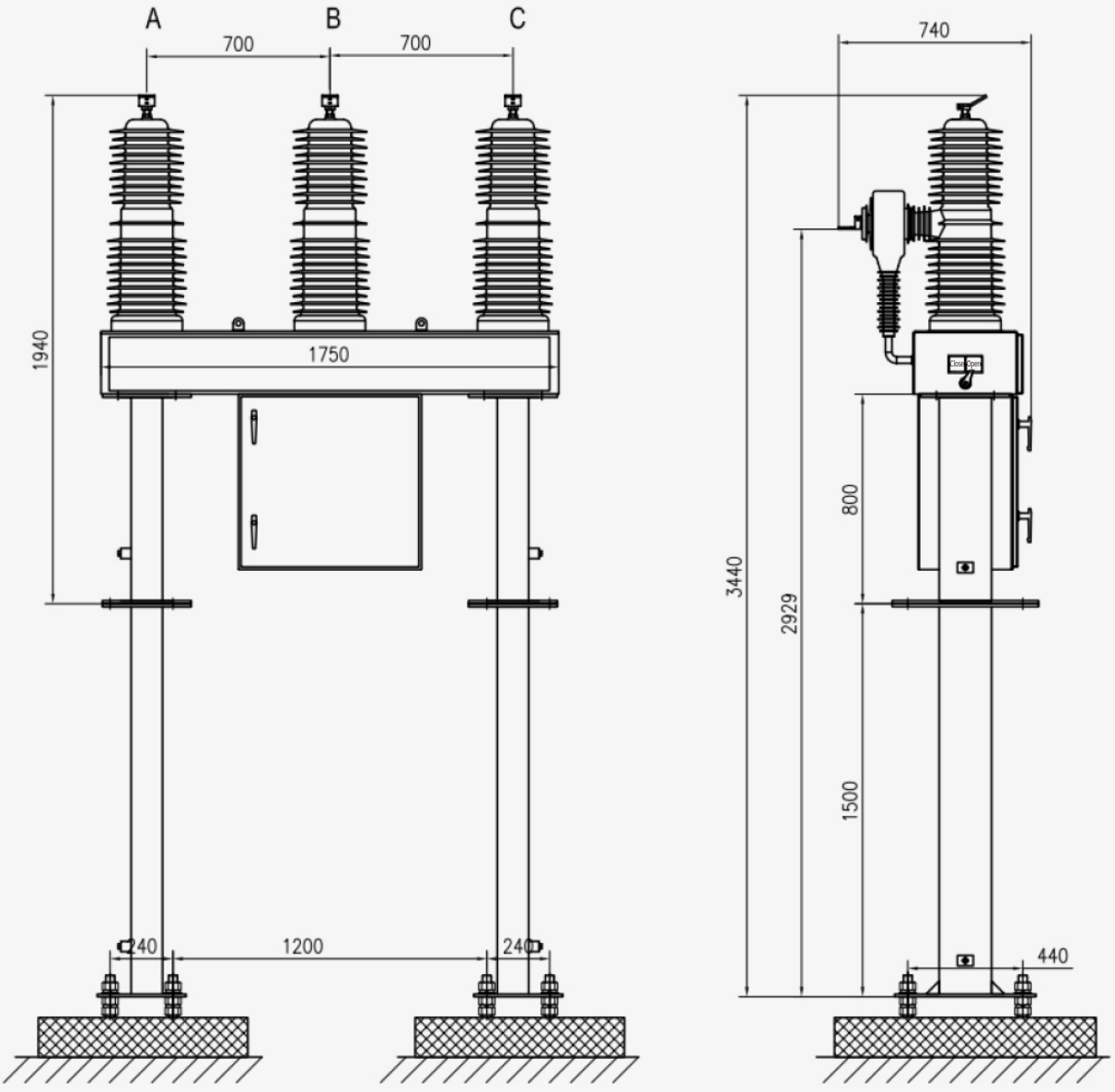


■ The upper and lower wiring terminals of the circuit breaker are made of copper, and the dimensions are as follows:



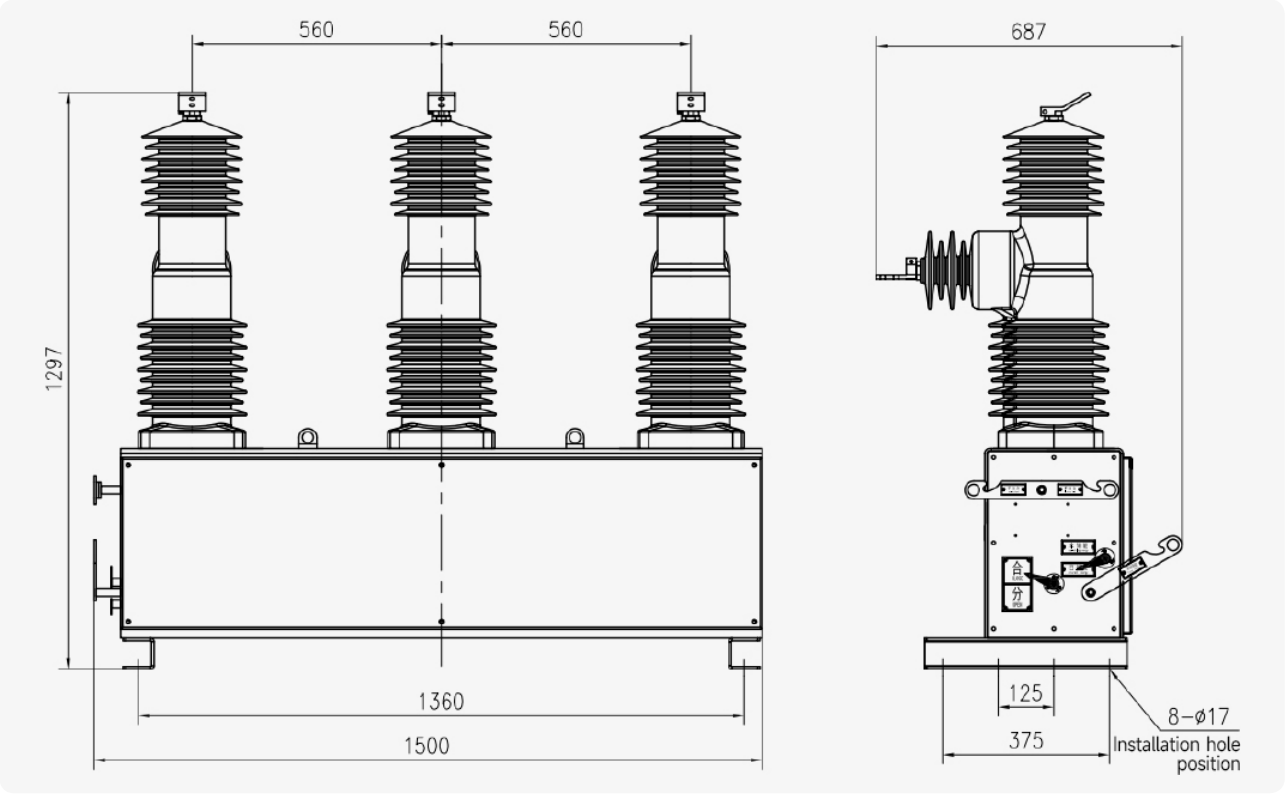
■ **Accurate and professional installation is a basic condition to ensure trouble-free operation of the product.**

The altitude is 1000 meters, 2000 meters, and the basic installation dimensions are the same standard.
The following installation methods are recommended:

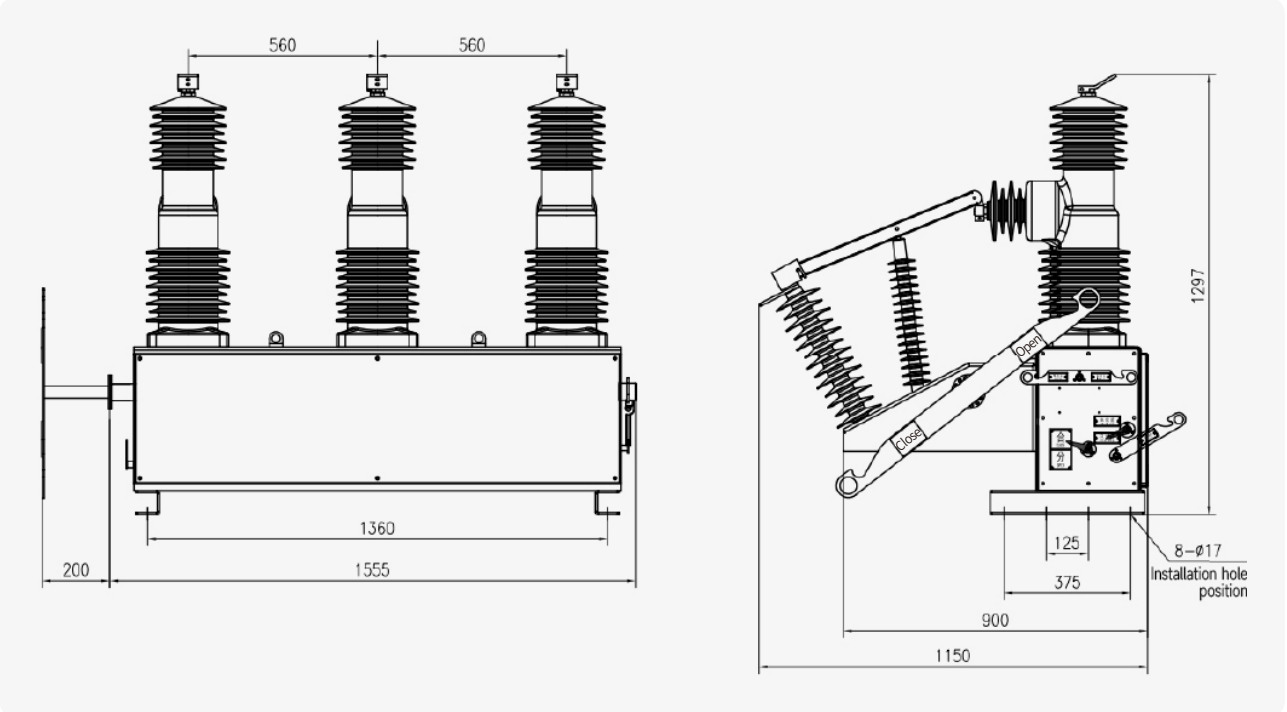


ZW7E-40.5 type, installation dimension diagram

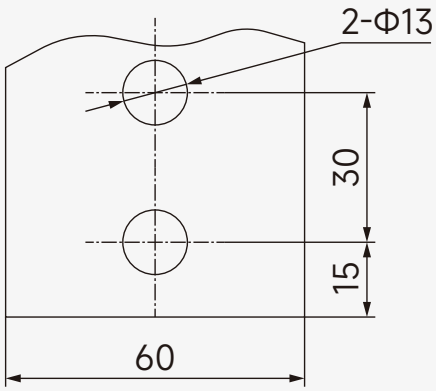
■ At an altitude of 1000 meters and an altitude of 2000 meters, the overall dimensions of the circuit breaker are as follows:



■ At an altitude of 1000 meters and an altitude of 2000 meters, the external dimensions of the circuit breaker with isolation are as follows:



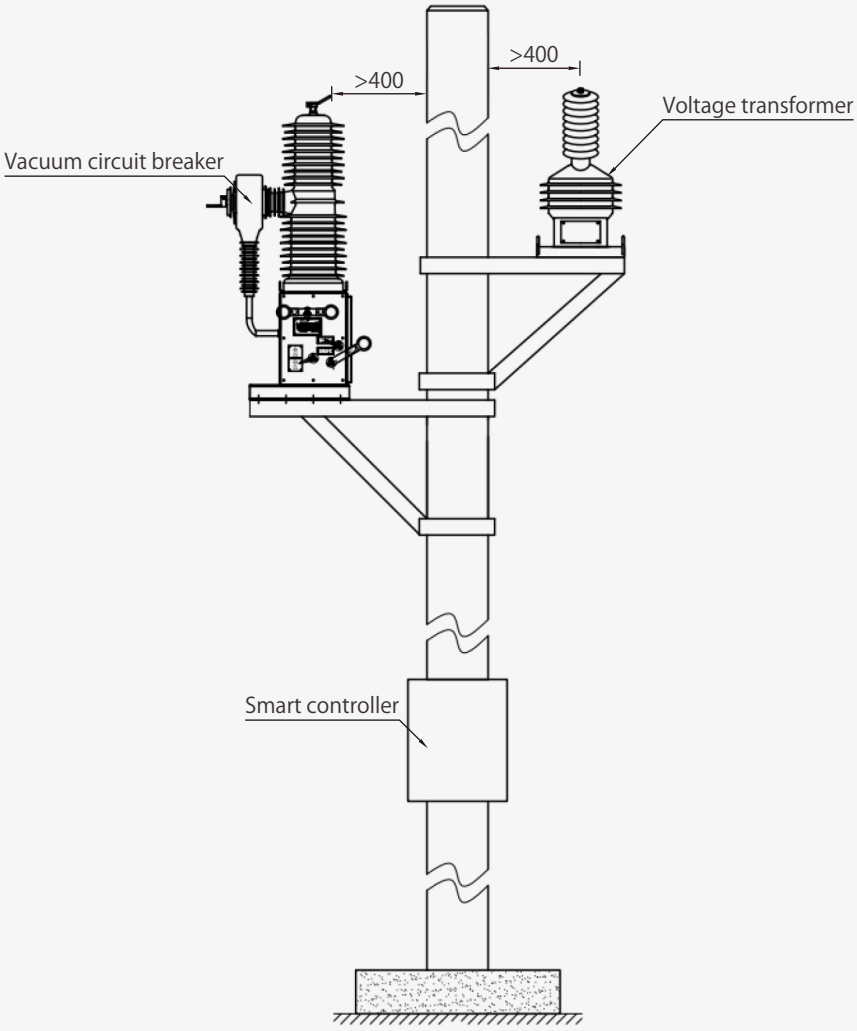
■ **The upper and lower wiring terminals of the circuit breaker are made of copper, and the dimensions are as follows:**



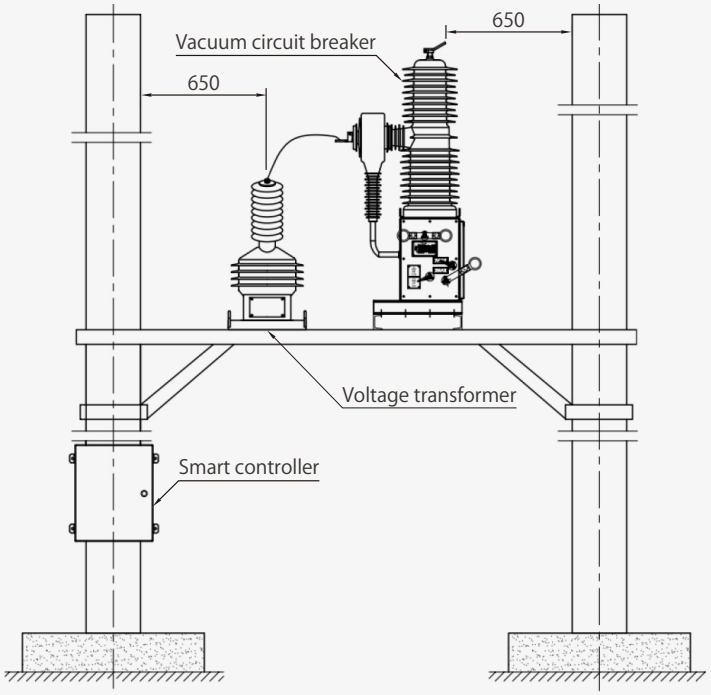
■ **Accurate and professional installation is a basic condition to ensure trouble-free operation of the product.**

The altitude is 1000 meters, 2000 meters, and the basic installation dimensions are the same standard.
The following installation methods are recommended:

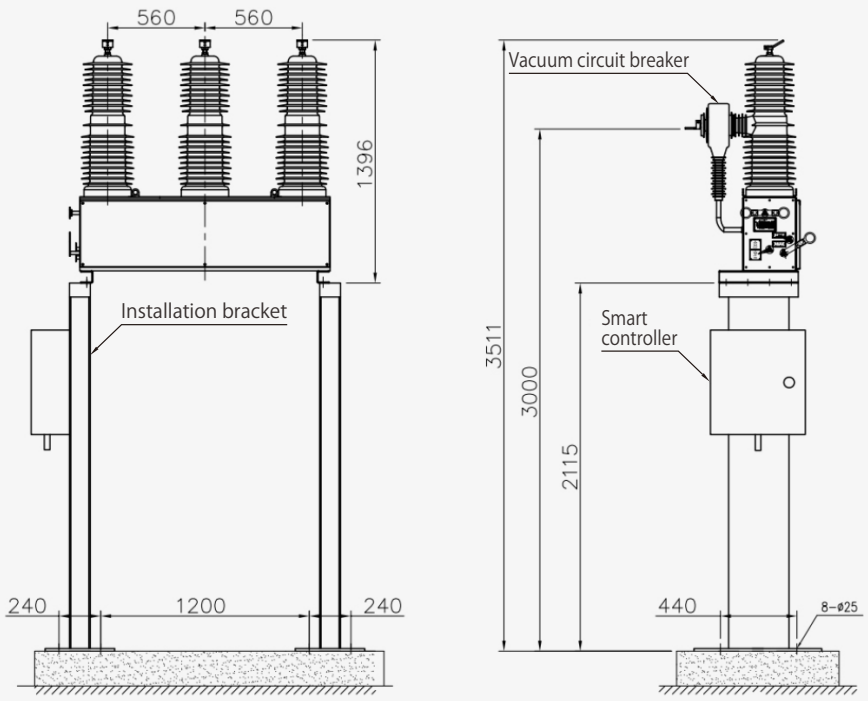
a. Single pole installation



b.Double pole installation



c.Floor installation



Adjustment

Before leaving the factory, all technical parameters of this product have been adjusted in strict accordance with national standards and technical specifications. No adjustment is required before installation and use.

8. Delivery conditions

Factory-assembled products must undergo integrity inspection before leaving the factory, and strict factory tests shall be carried out in accordance with the national standard GB1984, product technical conditions, and ordering technical specifications to ensure the correctness of each product, structure and function.

9. Packing, loading, unloading, acceptance and storage

Packaging

Before packaging, you should carefully check the completeness of the random information, the correctness of the nameplate, the completeness of the accessories and spare parts, and the consistency of the project; fix all movable parts; check the appearance integrity, no paint, bumps, and clean;

Packaging method 1: Pack the product in a dry room, wrap the product in a bubble nylon bag, wrap the product as a whole, and wrap it tightly with adhesive tape; for protruding parts or fragile parts, use soft materials, such as plastic foam, for protection. The inspection qualification standard is the packaging method, which meets the requirements of rainproof, anti-collision and shockproof effects;

Packing method 2: Pack the product in a dry room, wrap the product in a nylon bag, wrap the product as a whole, and wrap it tightly with adhesive tape; for the protruding parts or fragile parts, use soft materials, such as plastic foam, for protection; The product is placed in a wooden box and fixed, and the inspection standard is the packaging method, which meets the requirements of rainproof, anti-collision and shockproof effects;

Then, fix the relevant technical parameter table of the product, the project name and the receiving contact information on the conspicuous place outside the packing box.

Handling, loading and unloading trucks, transportation

When transporting, please note the following:

All movable parts must be fixed to prevent sudden shaking of the movable parts, which may cause product damage and personal safety;

When loading and unloading trucks, please note the following:

- Only use the following tools: forklift, crane or crane.
- Before loading and unloading the truck, all movable parts must be fixed to prevent safety accidents caused by sudden shaking of the movable parts.
- It is recommended to use a forklift during the loading and unloading process. When using a forklift for loading and unloading, it should be inserted in the direction shown in the figure and tied with a rope to avoid overturning.
- When lifting, the hook should not be hooked on the product body, but should be hooked in the lifting hole marked with the lifting mark on the product. The length of the hoisting rope should not be less than 5m. Try to use a soft rope, and fill the space between the hoisting rope and the product with soft materials, such as foam plastic, so as not to damage the surface of the product.
- Because the center of gravity of the product is eccentric to the upper rear, it may tilt backwards during fork loading and lifting operations.
- The operation speed should be slow to prevent the possibility of overturning, and attention should be paid and supported or tied.

When transporting, please note the following:

The whole unit must be packaged first in accordance with the requirements of the relevant provisions of [Packaging], and then fixed on the transportation vehicle.

In the process of handling, loading and unloading vehicles, and transportation, do not turn or tilt; take effective anti-vibration measures to avoid violent impact when landing; avoid non-focusing points as stressed parts, forced operations and other behaviors that endanger product quality.

Acceptance

When receiving the goods, the consignee should check the batch, quantity, model specification, and whether the outer packing box is damaged in accordance with the **"Delivery List"**;

If the packing box is damaged, the delivery party and the receiver should take further steps to unpack the product in person. If the product is damaged or there is a discrepancy in the batch, quantity, model, etc., the manufacturer should be notified immediately;

During normal unpacking and acceptance, carry out it in a dry room, check the **"Packing List"** to check whether the information and accessories are complete. Check for obvious cracks or damage on the outer surface. Check the nameplate technical parameters, random documents, etc., whether they meet the ordering technical requirements. Check whether the product is damp or has other adverse effects;

Manual trial operation, check whether each functional unit is normal;

This product has passed strict factory entry, process, factory test, and factory inspection, and meets the corresponding standards. If routine inspection and testing before putting into operation is required, loop resistance, power frequency withstand voltage, and mechanical operation should be carried out in accordance with relevant regulations. Waiting for handover test;

If the goods are found to be severely damaged, the supplier shall be notified in time, records shall be made, and photographs shall be taken if necessary.

Storage

This product should be stored in an indoor environment, with the characteristics of dry, ventilated, moisture-proof, shock-proof and harmful gas erosion; long-term storage, lubricating oil should be applied to the transmission part, and regularly check whether the environment meets the requirements;

The storage period of the vacuum interrupter is 20 years.

10. Use, operate, put into operation

Preparation before use

Check the product for damage or any other harmful environmental impact. If this phenomenon occurs, it should be eliminated in advance to restore the normal working conditions of the product;

Clean the surface of the insulating parts of the product with clean silk cloth and absolute ethanol. These dirt have great hidden dangers to the insulation performance of the product;

Check whether the product's operating voltage, operating mode, interface size, rated voltage, rated current, rated short-circuit breaking current, product configuration, etc. are consistent with the requirements of the project;

Check whether the accessories of the product are complete and meet the actual needs of the project.

Field test

Power frequency withstand voltage test:
The test voltage between the poles, the ground and the fracture is 95kV, and the test time is 1min;
There is no breakdown or flashover during this period;

Insulation test of auxiliary circuit:
The test voltage is to the ground: 2000V/1min, during which there is no breakdown or flashover phenomenon;
The electronic components in the auxiliary circuit shall not be subjected to the withstand voltage test, and shall be removed or isolation measures shall be taken during the withstand voltage test;
The insulation test of the energy storage motor adopts the insulation test voltage of 1600V1min to carry out the withstand voltage test;

Main loop resistance measurement:
Use loop resistance tester to measure, in line with the specifications of technical parameters;

Mechanical characteristic test:
Operate the circuit breaker under the rated operating voltage and measure it with a mechanical characteristic tester, and its value meets the relevant regulations.

Mechanical operation test:
Under the specified operating voltage, continuously open and close 10 times each, and its action should be normal;
Operate the product manually, and operate the closing and opening buttons 5 times each, and the action should be normal;
In the closed state, at 30% of the rated operating voltage, three consecutive opening operations should not be possible.

Setting of protection setting

Line protection tests such as inrush current, over current, and quick disconnection shall be carried out, and other parameters shall be set.

11. Maintenance

The maintenance of the product should be carried out under normal conditions when the equipment is in the state of power failure, discharge, isolation, grounding, opening, and no energy storage;

Maintenance and maintenance must be completed by professionals who are familiar with various safety operating procedures and understand the product performance;

Necessary records should be made for maintenance and upkeep;

When performing maintenance work without power failure, make sure that there is no electric shock or other hazards;

The products produced by our company have the characteristics of simple structure and durability, so they have a long service life. Under normal conditions of use, there is very little maintenance work within the service life. However, due to the difference in the use environment and its impact on the service life of the product, the following necessary inspections and maintenance work are still required;

Depending on the working environment, the appearance of the product body should be inspected regularly;

At the same time, check the contamination, moisture and corrosion on the surface of the equipment;

Use a dry cloth to clean the dust on the surface of the insulation, and then use a silk cloth moistened with a cleaning agent to wipe off other contaminants, but pay attention to the cleaning agent used, which can be applied to paint or synthetic insulation materials;

If the product is used in occasions where it will not be operated for a long time, the circuit breaker should be properly closed and opened every year, and the rotating friction parts should be lubricated.

Common faults and troubleshooting methods

No.	Failure phenomenon	Cause of issue	Method of exclusion
1	The circuit breaker cannot be closed	The circuit breaker is already in the closed state	Open breaker
		No power for closing circuit	Check whether the power connection is correct and plug in the power
		In manual operation, press the manual opening operation button by mistake	Press the closing operation button
		The transfer switch is in the "Local" position	Turn the switch in the "Remote" position
2	The circuit breaker cannot be opened	The circuit breaker is already in the open state	Close breaker
		No power in the opening circuit	Check whether the power connection is correct and plug in the power
		Wrongly press the manual closing operation button during manual operation	Press the opening operation button
		The transfer switch is in the "Local" position	Turn the switch in the “Remote” position

12. Document

- Certificate of Compliance;
- Factory inspection report;
- Installation instructions;
- Electrical schematic diagram;
- Installation base map;
- Packing list, including tools, accessories, spare parts, spare parts, etc.;
- Quality information feedback sheet.

13. Ordering notice

- Product name, model, specification, quantity;
- Mechanism operating voltage;
- The transformation ratio, accuracy, capacity and quantity of current transformers;
- The name, specification and quantity of accessories, spare parts and spare parts;
- Material
- Electrical schematic diagram;
- Installation base map;
- Use environmental conditions;

14. Ordering technical specifications

No.	Name	Data 1	Data 2	Data 3	Data 4	Remarks
1	Order qty.	1 set ☐	2 sets ☐	3 sets ☐	Other qty. ☐	
2	Structure type	ZW7A-40.5 ☐ Suit for 5000m	ZW7B-40.5 ☐ Suit for 2000m	ZW7C-40.5 ☐ Suit for 2000m	ZW7D-40.5 ☐ Suit for 2000m, current 1250A	ZW7E-40.5 ☐ Suit for 2000m, current 1250A
3	Rated voltage	35kV ☐				
4	Rated current	630A ☐	1250A ☐	1600A ☐	2000A ☐	2500A ☐
5	Rated short-circuit breaking current	20kA ☐	25kA ☐	31.5kA ☐		
6	Current transformer	Without ☐	With ☐	Built-in ☐	External ☐	Demand, structure
		25-50-75/5	50-100-150/5	100-200-300/5	200-400-600/5	Typical parameters
		0.2S	0.5	10P10	5P20	
		10VA	20VA	25VA	10VA	
						Ratio
						Accuracy
						Capacity
7	Voltage transformer	Without ☐	With ☐	Qty. 1 set ☐	Qty. 2 sets ☐	Demand
		35/0.1kV ☐	35/0.22kV ☐	35/0.1/0.22kV ☐		Ratio
		0.2 ☐	0.5 ☐	3 ☐	0.5/3 ☐	Accuracy
						Capacity
8	FTU controller	Without ☐	With ☐	UPS ☐	GSM message ☐	Back-end networking ☐
9	Meter box	Without ☐	With ☐	1 gauge ☐	2 gauge ☐	
10	Outer box material	Steel plate coated ☐	201 coated ☐	304 coated ☐		
		201 ☐	304 ☐			
11	Switch mounting bracket	Without ☐	With ☐			Demand
		Height 1.2m ☐	Height 1.5m ☐	Other height ☐		Height
		Powder coated ☐	Galvanized ☐			Surface treatment
12	Abnormal use conditions	2000m above sea level ☐	3000m above sea level ☐	4000m above sea level ☐	5000m above sea level ☐	The default altitude is 1000m
		Heavy pollution class III ☐	Seriously dirty class IV ☐	Severe cold ☐	Severe condensation ☐	
		Other environmental conditions ☐				Specific description
13	Operating mechanism	Spring mechanism ☐	Permanent magnet mechanism ☐			Default spring mechanism
14	Motor energy storage operating voltage	DC110V ☐	DC220V ☐	AC110V ☐	AC220V ☐	Spring mechanism
15	Opening and closing operating voltage	DC110V ☐	DC220V ☐	AC110V ☐	AC220V ☐	Spring mechanism
16	Installation works	Manufacturer standard ☐	Provided by the project unit ☐			
17	Electrical schematic diagram	Manufacturer standard ☐	Provided by the project unit ☐			
18	Spare parts	None ☐	Yes ☐			Specification type quantity
19	Other special requirements					

Remarks:

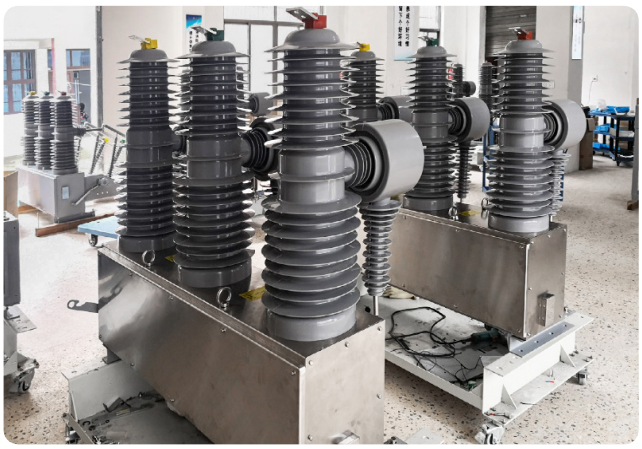
Annex. 1 Controller FTU instructions

Instructions for use of controller FTU for details, please refer to **"Single Installation Manual"**.

Annex. 2 Similar products













Annex. 3 Some use cases





E5.0202ME-202301

Zhejiang Greenpower Electric Co., Ltd
Zhejiang GreenPower I&E Co., Ltd
Professional Power Equipment Provider

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